

Large Language Models and Argumentation

A corpus linguistics study of the use of linking adverbials in
argumentative essays generated by Large Language Models

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Introduction

- Large language models (LLMs) have made significant advancements in generating human-like text, exhibiting impressive fluency and coherence.
- Companies are increasingly integrating LLMs into their business processes and many face a critical decision when selecting an appropriate LLM, particularly whether to opt for open-source or proprietary (closed) models. This choice can significantly impact factors such as cost, customization, and data privacy.
- However, LLMs are fundamentally statistical models designed to predict and generate text based on patterns they learn from data, which raises questions about their abilities in tasks that involve logical thinking.
- For these reasons, argumentative writing can be challenging for LLMs, as it aims to persuade an audience by presenting reasons and evidence to support a particular claim or position.



Introduction

- The aim of this research is to compare human-written argumentative essays with essays generated by LLMs and to analyze differences in argumentative writing across various LLMs.
- The research focuses on the use of linking adverbials, which play a crucial role in establishing logical connections and cohesion within argumentative essays.
- The research follows a corpus linguistics approach to ensure empirical, objective, and quantitative results. A qualitative approach is used to address specific questions.
- Code and data available at github.com/tonazzog/LLM_Argumentation/



Agenda

Background

Relevance and limitations

Research questions

Methodology

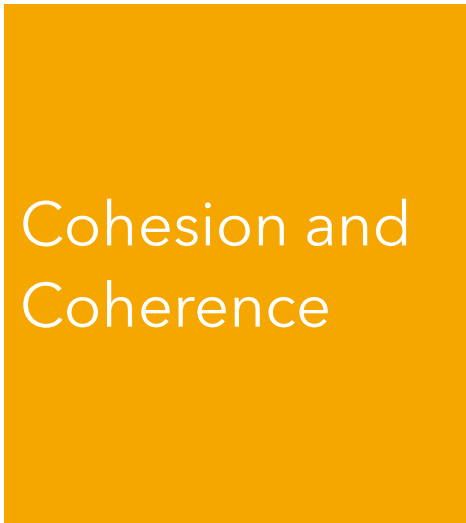
Results

Discussion

Conclusion



Background



Cohesion and
Coherence

Cohesion and Coherence

In the study of discourse in general, and of argumentative discourse in particular, the concepts of coherence and cohesion are fundamental to understanding how texts are structured and how meaning is conveyed [Halliday and Hasan, 1976]

Coherence

- refers to the overall sense of unity and logical flow within a text
- abstract quality that emerges when all parts of a text work together to create a meaningful whole
- guides the reader through the argument or narrative in a way that is clear and easy to follow
- achieved through the logical arrangement of ideas and the clear development of themes
- uses implicit connections that allow readers to understand the relationships between different parts of the text

Cohesion and Coherence

Cohesion

- refers to the linguistic devices and grammatical structures that link sentences and parts of the text together
- it is about the surface connections between words, phrases, and clauses that create a sense of unity
- is concerned with the formal links between elements in the text, ensuring that it is linguistically connected
- occurs where the interpretation of some element in the discourse is dependent on that of another
- is achieved through the use of **cohesive devices**

Cohesive Devices

[Halliday and Hasan, 1976] identify several types of cohesive devices, including:

- **Reference:** using pronouns or other referring expressions to link parts of a text ("John went to the store. *He* bought some milk")
- **Substitution:** replacing a word or phrase with another word to avoid repetition ("I need a pen. Do you have *one*?")
- **Ellipsis:** omitting part of a sentence because it is understood from the context. "I'll take the first bus, and you [take] the second."
- **Conjunction:** using linking words (such as "and", "but", "therefore") to connect sentences or parts of sentences.
- **Lexical Cohesion:** the repetition of words, synonyms, or related words to create connections between parts of the text. "The *cat* sat on the mat. The *feline* seemed content."

Conjunctions

Conjunctions are crucial for creating cohesion in a text because they explicitly signal the relationships between different parts of a text, guiding the reader through the logical structure of the argument. [Halliday and Hasan, 1976] categorize conjunctions into several types based on the kind of relationship they establish between ideas:

- **Additive** conjunctions add information or ideas. Examples: and, furthermore, moreover, in addition, besides ("She enjoys reading and writing.")
- **Adversative** conjunctions introduce a contrast or opposition between ideas. Examples: but, however, on the other hand, nevertheless, yet. ("The weather was terrible, but we decided to go hiking anyway.")
- **Causal** conjunctions express a cause-and-effect relationship. Examples: because, therefore, so, thus, consequently, hence. ("He studied hard for the exam, so he passed with high marks.")
- **Temporal** conjunctions indicate the timing or sequence of events. Examples: then, next, after that, before, meanwhile, finally. ("First, we went to the museum. Then, we had lunch.")

Linking Adverbials

Biber uses the term "**linking adverbials**" to describe words or phrases that function similarly to Halliday's conjunctions [Biber et al., 1999, pp 765, 875]:

- "those adverbials that serve to connect two stretches of discourse"
- they "make explicit the relationship between two units of discourse" and are therefore "important devices for creating textual cohesion"
- their main function is "to state the speaker/writer's perception of the relationship between two units of discourse" and "to make semantic connections between spans of discourse of varying length"

Biber's classification of linking adverbials is based on the corpus analysis of how these adverbials function in actual usage.

Comparison between conjunctions and linking adverbials

Analogies:

- both classifications deal with how ideas are connected in discourse, with significant analogies in their functional categories (e.g. additive, adversative, ...).

Differences:

- Biber uses the term "linking adverbials" to emphasize their role as standalone elements (usually at the beginning of a sentence or clause), while Halliday talks about "conjunctions" more broadly, including within-clause connections.
- Biber addresses the issue of the meaning of linking adverbials: they should be viewed as meaning markers that contribute to discourse cohesion by indicating how the reader should interpret information that follows in light of what has preceded it.

Classification of Linking Adverbials

Category	Function	Examples
Enumeration and Addition	- enumeration of pieces of information in an order chosen by the speaker/writer - addition of items of discourse to one another	enumeration: first(ly), second(ly), next, first of all addition: in addition, similarly, also, furthermore, likewise, moreover
Summation	show that the unit of discourse is intended to conclude or sum up the information in the preceding discourse	in sum, to conclude, all in all, in conclusion, overall, to summarize
Apposition	show that the second unit of text is to be treated either as equivalent to or included in the preceding unit; the second unit can be a - restatement of the first, reformulating the information it expressing some way or stating it in more explicit terms - an example, and it is presented as information that is in some sense included in, rather than exactly equivalent to, the previous text	which is to say, in other words, that is, e.g., for example, for instance, namely, specifically

Classification of Linking Adverbials

Category	Function	Examples
Result/ inference	show that the second unit of discourse states the result or consequence - either logical or practical - of the preceding discourse	result: therefore, consequently, thus, so Inference: then
Contrast/ concession	• contrast: can mark incompatibility between information in different discourse units • concession: can mark concessive relationship, showing that the subsequent discourse expresses some reservation about the idea in the preceding clause	contrast: on the other hand, in contrast, alternatively concession: though, anyway, anyhow, besides, nevertheless, after all
Transition	mark the insertion of an item that does not follow directly from the previous discourse; the new information is not incompatible with what it is linked to, but rather it is only loosely connected, or unconnected.	by the way, incidentally

The interactive role of linking adverbials

Linking adverbials are an important part of **Hyland's** [Hyland, 2018] theory of **metadiscourse**. Metadiscourse is the language that signals the writer's or speaker's attitude towards the content, their relationship with the audience, and the organization of the discourse. It acts as a guide to the reader or listener, helping them navigate the text and understand the intended meaning. Hyland distinguishes between

- Interactive metadiscourse: concerns the ways in which writers organize their discourse to make it coherent and easy for readers to follow. It includes elements like transitions, frame markers, endophoric markers, evidentials, and code glosses.
- Interactional metadiscourse: involves engaging with the reader, including the writer's stance, engagement with the audience, and how the text reflects the relationship between writer and reader (includes hedges, boosters, attitude markers, etc.)

Linking adverbials fall within the category of **interactive** metadiscourse. In this framework, linking adverbials are key tools that writers use to manage the flow of ideas and to interact with the readers, ensuring that they can follow the logical progression of the text.



Relevance and Limitations



Limitations

Cohesion does not always imply coherence...

The aim of this research is to compare the use of linking adverbials in human-written argumentative essays with essays generated by LLMs. These cohesive ties help to create a connected and unified text, which is a fundamental aspect of coherence

...but cohesion does not always imply coherence: a text can be cohesive without being coherent if the ideas don't make logical sense together, or if the reader cannot easily follow the intended meaning. For instance, a text might use cohesive devices but still be confusing or illogical due to poorly structured arguments or irrelevant content.

Some studies (i.e. [Appel & Szeib, 2018]) have shown that L2 learners tend to overuse linking adverbials to “to impose surface logicity on a piece of writing where no deep logicity exists”.

Limitations

... but there is a correlation

Other studies have shown that there is a correlation between the use of cohesive ties and coherence.

[Weiss et al., 2019] observe the correlation between measures of lexical, syntactic and discourse complexities to the essay gradings in high-school examinations.

[Somasundaran et al. 2018] use discourse markers to evaluate the story-telling aspect of student writing and show that discourse markers can be used to measure the coherence quality of a text.

In this research we will consider linking adverbials as an indicator of coherence in argumentation.

Relevance

We know little about the linguistic and dialogical differences of argumentation between AI and human written texts

[Tudino & Qin, 2023] write that “the inquiry into AI and ChatGPT’s role in academia predominantly centers on ethical and authenticity concerns”.

Significant portion of the literature is dedicated to the ethical issues emerging from the model’s utilization and the threats associated with large language models and their impact on academic works’ authenticity and credibility (i.e. [Sardinha, 2024] compares human written and LLM generated text focusing on authorship problems).

Considerable attention is devoted to content (i.e. [Hinton & Wagemans, 2023] study the quality of argumentative texts produced by GPT-3), whereas “form” (intended as the AI-generated-texts linguistic resemblance to human writing) remains largely unexplored.

This suggests the need for further in-depth investigation into the linguistic characteristics and patterns of AI-generated academic prose to determine whether ChatGPT effectively emulates human writing also in terms of language choices.

Related work

Two papers investigate into the linguistic characteristics and patterns of AI-generated texts.

[Tudino & Qin, 2023] analyze the differences in terms of parts of speech, subordinating conjunctions, 3-gram and 4-gram phraseologies between ChatGPT-generated academic texts in comparison to human-authored texts in the field of social sciences.

They conclude that ChatGPT can produce academic-style texts, but falls short in terms of linguistic complexity, coherence, and the appropriateness of formal academic language.

The authors note that research focuses on social sciences, and the result may not be generalizable to other academic fields.

Coherence discourse markers are not analyzed, and text are generated only ChatGPT-3.5 (so the research is already outdated).

Related work

[Herbold et al. 2023] compare AI-generated argumentative essays with essays written by students. They analyze nominalization and sentence complexity, modal and epistemic constructions, lexical diversity and the use of linking discourse markers.

They combine human scoring and quantitative analysis and conclude that ChatGPT-generated essays were generally rated higher in quality than human-written. ChatGPT-4 outperformed ChatGPT-3.5 in most areas.

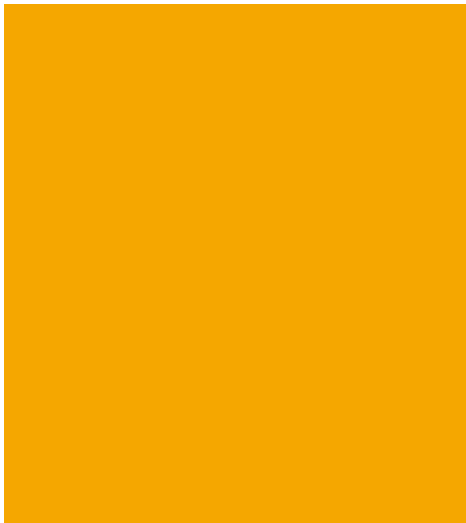
They employ the Penn Discourse Treebank list of linking [discourse markers](#), which does not include all of Biber's adverbials (only single word markers are included) and discourse markers are not categorized according to their semantic function.

They find that "While there is no difference in the use of discourse markers between humans and ChatGPT-3, *ChatGPT-4 uses significantly fewer discourse markers*" and conclude that "We believe that this might be due to the rigid structure of the generated essays: *instead of using discourse markers, the AI models provide a clear logical structure by separating the different arguments into paragraphs*, thereby reducing the need for discourse markers."

They compare ChatGPT-3.5 with ChatGPT-4, other LLMs are not considered.



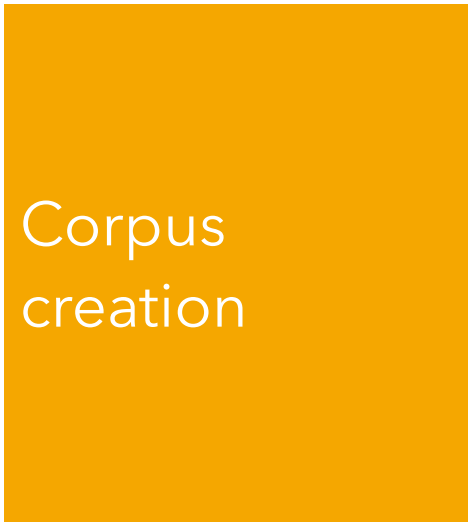
Research questions



- Q1 - How do patterns of linking adverbial usage differ between student-written and LLM-generated argumentative essays?
 - Q2 - What variations exist in linking adverbial use among different LLMs?
 - Q3 - What do these differences imply about the argumentative writing of LLMs?
-



Methodology



Corpus
creation

Options for reference corpus

Corpus of Argument-Annotated Persuasive Essays

- [Stab & Gurevyc, 2017]
- contains argumentative essays annotated for various argumentative structures
- the lengths of the essays vary significantly, some consist of only two paragraphs

MICUPS Corpus

- Michigan Corpus of Upper-Level Student Papers
- essays often cover highly specialized topics across various fields (economics, philosophy...) and include extensive notes and references
- essays can be very long, with some extending to nearly 20,000 words

PERSUADE Corpus

- Persuasive Essays for Rating, Selecting, and Understanding Argumentative and Discourse Elements [Crossley et al., 2022]
- argumentative essays written by students in grades 6 through 12; essays are scored
- the average number of words of the essays which obtained the higher scores (5 and 6) is around 600



Selection of reference corpus

As a preliminary step, some LLMs were prompted to write an argumentative essay.

It was observed these essays tend to have an average length of approximately 400-500 words.

They typically follow a standardized structure, consisting of an introduction, three argumentative body paragraphs, and a conclusion.

The PERSUADE corpus was selected as the most appropriate reference corpus for comparison with the LLM-generated essays, due to its comparable essay length and focus on argumentative discourse on generic topics.

The PERSUADE corpus

The PERSUADE corpus is an open-source corpus of over 25,000 essays written by students in grades 6 through 12. It comprises two types of essays: source-based and independent essays.

- The source-based writing requires the student to refer to a text; essays were written from seven writing prompts and related sources. For two of the prompts, the source text could not be retrieved and only five source-based prompts were used.
- The independent set reflects writing where background knowledge of the topic was not a requirement, and no sources were required to produce the texts. The independent essays were derived from eight writing prompts.

The corpus includes holistic [scores](#) of writing quality for each essay. The scores use a scale between 1 (minimum) and 6 (maximum).

The PERSUADE corpus

Example of source-based essays

Driverless cars are coming: "In the article "Driverless Cars are Coming," the author presents both positive and negative aspects of driverless cars. Using details from the article, create an argument for or against the development of these cars. Be sure to include: your position on driverless cars; appropriate details from the article that support your position; an introduction, a body, and a conclusion to your argumentative essay."

Example of independent essays

Phones and driving: "Today the majority of humans own and operate cell phones on a daily basis. In essay form, explain if drivers should or should not be able to use cell phones in any capacity while operating a vehicle."

The PERSUADE corpus

Source-based essays topics

A Cowboy Who Rode the Waves
Car-free Cities
Does the electoral college work?
Driverless cars are coming
The Face on Mars

Independent essay topics

Cell phones at school, Community service,
Distance learning, Summer projects,
Grades for extracurricular activities
Mandatory extracurricular activities
Phones and driving, Seeking multiple opinions

Corpus creation

78 essays (6 for each topic) were randomly chosen from the PERSUADE corpus among the essays that were rated as good examples of student writing (with holistic scores greater than 5)

The corpus of LLM written essays

- Six large LLMs were tasked with generating argumentative essays on each of the 13 topics from the PERSUADE corpus.
- Three closed LLMs (ChatGPT, Claude and Gemini) and three open LLMs (Llama, Mistral, Gemma) were included.
- Objective of this research was to test LLMs “out of the box”, with minimal prompting strategy (zero shot prompting). The following prompt was used:

“Your task is to write an argumentative essay on the following topic. Your aim is to persuade the reader. You utilize critical thinking and logical reasoning in your argument to ensure discourse coherence and cohesion. You make extensive use of complex, compound sentences. You do NOT use lists or bullet points.

Topic: {one of the 13 essay topics}”

The corpus of LLM written essays

- The temperature parameter of LLMs was set to 0.7 (temperatures between 0.5 and 1 should balance coherence with creativity and diversity).
- For Claude and Llama interaction was via chat interface (Claude APIs require VAT number for registration, Llama3 APIs were not available in Italy).
- For the other models, interaction was via API calls using [LangChain](#) (Python framework that abstracts from LLMs differences and allows uniform interaction).

Python function that asks a LLM to generate an argumentative essay for each topic and saves the answer to a text file

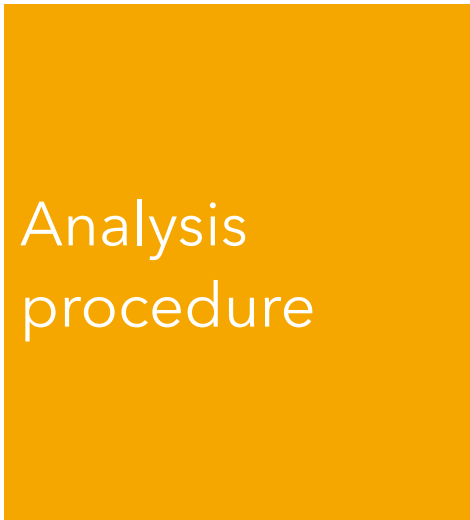
```
def invoke_llm(llm, name):
    for topic in topics:
        file_name = name + "_" + topic[0] + ".txt"
        prompt = ChatPromptTemplate.from_template("""
            Your task is to write an argumentative essay on the following topic. Your aim is to persuade the reader.
            You utilize critical thinking and logical reasoning in your argument to ensure discourse coherence and cohesion.
            You make extensive use of complex, compound sentences. You do NOT use lists or bullet points.
            Topic: {topic}
        """)
        output_parser = StrOutputParser()
        chain = prompt | llm | output_parser
        answer = chain.invoke({"topic": topic[1]})
        answer = clean_text(answer)
        with open(file_name, "w", encoding='utf-8') as f:
            f.write(answer)
```

LLM	Company	Version	Size (parameters)	Open	Interaction
 ChatGPT	OpenAI	Gpt-4	1.75 trillion (*)	No	API Key & LangChain
 Gemini	Google	Gemini 1.5 pro	500 billion+ parameters (*)	No	API Key & LangChain
 Claude	Anthropic	Claude 3.5 Sonnet	not publicly disclosed	No	Chat
 Llama 3	Meta	Llama3	70 billion	Yes	Chat
 Gemma	Google	Gemma2	27 billion	Yes	API Key & LangChain
 LE CHAT MISTRAL	Mistral AI	Mistral Large	123 billion	Yes	API Key & LangChain

(*) Estimations, actual parameters are not publicly disclosed by the organization



Methodology



Analysis
procedure

Procedure for data analysis

Step 1

A list of linking adverbials was compiled based on Biber's classification. For each adverbial, the corresponding regular expression was specified for search purposes. For instance, "to summarize" was represented as "to summar[is]e" to account for both American and British spelling variations.

Step 2

A custom Python procedure was developed to identify occurrences of these adverbials within the text. This procedure utilized regular expressions and recorded several key pieces of information for each match: the specific adverbial found, its starting position in the text, as well as the surrounding left and right context.

Step 3

A manual check of the results was conducted using the AntConc software.

This step allowed for verification that the automated process hadn't missed any relevant instances.

Procedure for data analysis

Step 4

The results of the search were then saved in an Excel file.

A manual analysis of these results was performed to exclude any "false positives" that may have been captured by the automated process.

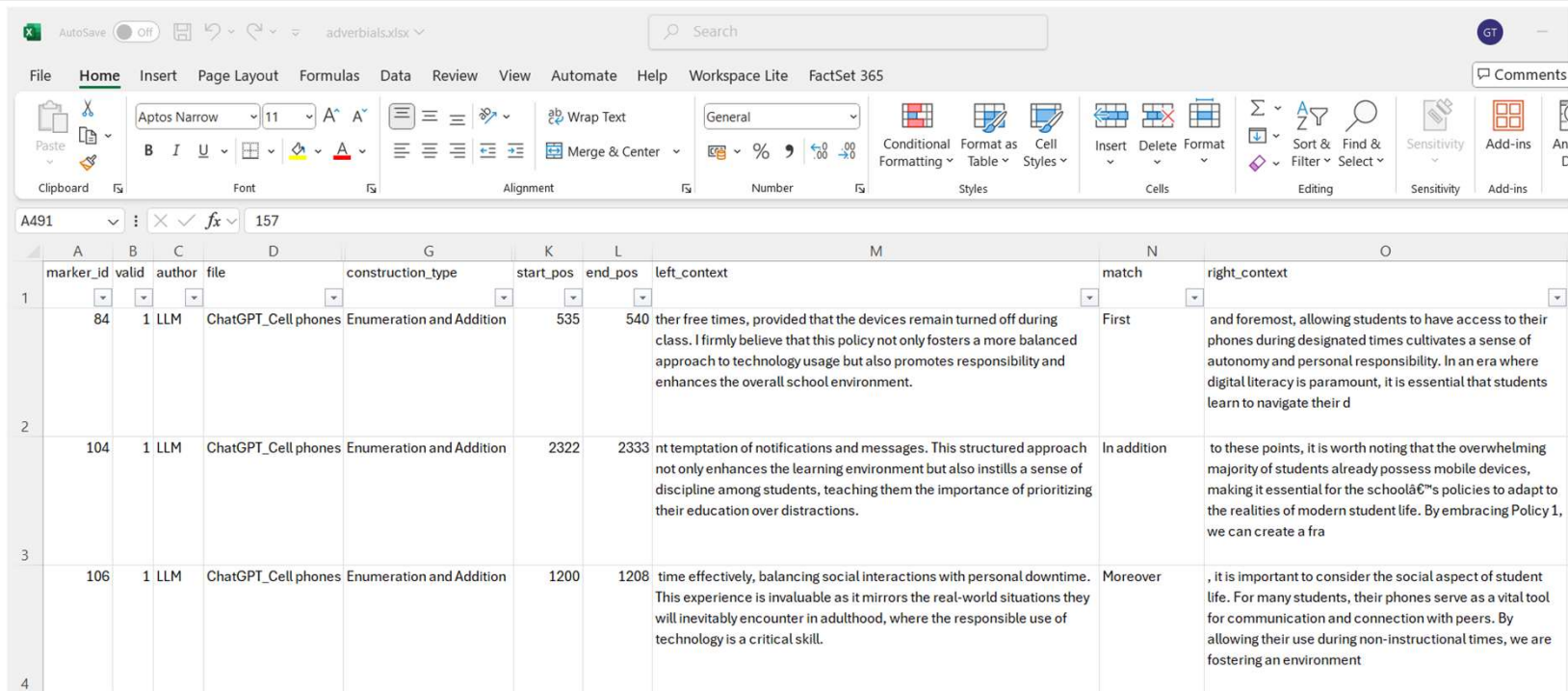
Step 5

The Stanford Stanza natural language processing library with the "en" pipeline was used to parse the text. This allowed for the computation of the number of tokens and sentences for each file. According to the documentation, its accuracy rates are 99.77% for tokenization and 92.20% for sentence segmentation

Step 6

Data analysis and visualization were conducted using SQL server, R programming language and ggplot2 library, and AntConc software.

The Excel file generated by the procedure



	A	B	C	D	G	K	L	M	N	O
	marker_id	valid	author	file	construction_type	start_pos	end_pos	left_context	match	right_context
1	84	1	LLM	ChatGPT_Cell phones	Enumeration and Addition	535	540	ther free times, provided that the devices remain turned off during class. I firmly believe that this policy not only fosters a more balanced approach to technology usage but also promotes responsibility and enhances the overall school environment.	First	and foremost, allowing students to have access to their phones during designated times cultivates a sense of autonomy and personal responsibility. In an era where digital literacy is paramount, it is essential that students learn to navigate their d
2	104	1	LLM	ChatGPT_Cell phones	Enumeration and Addition	2322	2333	nt temptation of notifications and messages. This structured approach not only enhances the learning environment but also instills a sense of discipline among students, teaching them the importance of prioritizing their education over distractions.	In addition	to these points, it is worth noting that the overwhelming majority of students already possess mobile devices, making it essential for the schoolâ€™s policies to adapt to the realities of modern student life. By embracing Policy 1, we can create a fra
3	106	1	LLM	ChatGPT_Cell phones	Enumeration and Addition	1200	1208	time effectively, balancing social interactions with personal downtime. This experience is invaluable as it mirrors the real-world situations they will inevitably encounter in adulthood, where the responsible use of technology is a critical skill.	Moreover	, it is important to consider the social aspect of student life. For many students, their phones serve as a vital tool for communication and connection with peers. By allowing their use during non-instructional times, we are fostering an environment
4										

Examples of manual analysis

	marker_id	valid	author	file	construction_type	start_pos	end_pos	left_context	match	right_context
1	84	1	LLM	Mistral_Summer proje	Enumeration and Addition	615	620	cultivate essential skills that extend beyond the confines of the traditional classroom. In the subsequent paragraphs, I shall elucidate the primary reasons underpinning my position and augment them with specific examples to substantiate my claim.	First	and foremost, student-designed summer projects empower young learners to take ownership of their education, fostering a sense of responsibility and intrinsic motivation. When students are entrusted with the task of designing their own projects, they
457	84									
594	84		student	2B3AA27C59CB_Seek	Enumeration and Addition	1083	1088	come out of your decision. Then, they can help you weigh each decision out and make your choice. Say you're deciding what classes you're taking in college. Your counselor and parents can help you decide whether you want to take chemistry or physics	first	year. If you take chemistry then you get more credits, but you like physics much more. Which one would you pick? The more beneficial class, or the more enjoyable class? That's where asking for help comes in handy. Furthermore, seeking multiple opin
1693	148	1	LLM	Gemini_A CowboyWh	Contrast/Concession	1849	1853	reward far more valuable than any material possession. The program is not without its challenges, as the open sea can be unpredictable and unforgiving. I myself experienced a close call, narrowly avoiding a dangerous fall into the churning Atlantic.	Yet,	It is in these moments of adversity that true character is forged, resilience is built, and a profound appreciation for the fragility of life takes root. Therefore, I implore you to seize this extraordinary opportunity, to step outside your comfort
	148		Student	01FCFE3F279B_Car-fi	Contrast/Concession	6439	6442	reach our destination. Safety improvements need to happen to make drivers have common sense and to not run over someone crossing the street. Resources are limited everywhere, and what we'll do after they've been gone? That answer has not been found	yet	. Saving time is our number-one priority, and we try to beat our records every time we go somewhere, whether it be using the same mode of transportation, or trying out a new one. Next time, be spontaneous for a day, and don't use a

"First" as linking adverbial vs adjective, "Yet" as linking adverbial vs time adverbial

Other examples: Rather: exclude "rather quiet", "rather tired", Similarly: exclude e.g. "men and women behave similarly"



Results



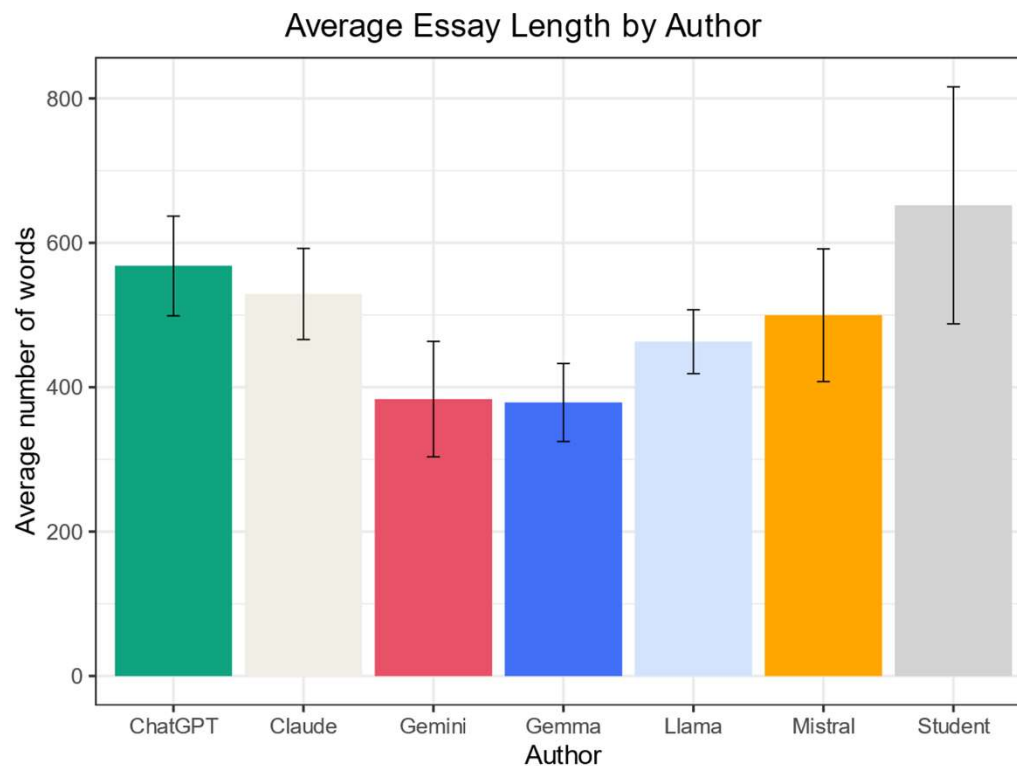
Corpus size and number of adverbials

The corpus of student-written essays contains more words and more linking adverbials.

The number and the proportion of false positives were higher in the student corpus.

	Students	LLM	Total
Number of files	78	78	156
Corpus size (number of tokens)	50849	36689	87538
Linking adverbials (retrieved by the procedure)	1152	729	1881
Valid linking adverbials (after false positive removal)	725	600	1325
False positive percentage	37%	18%	30%

Essay length



Essays by students are on average longer.

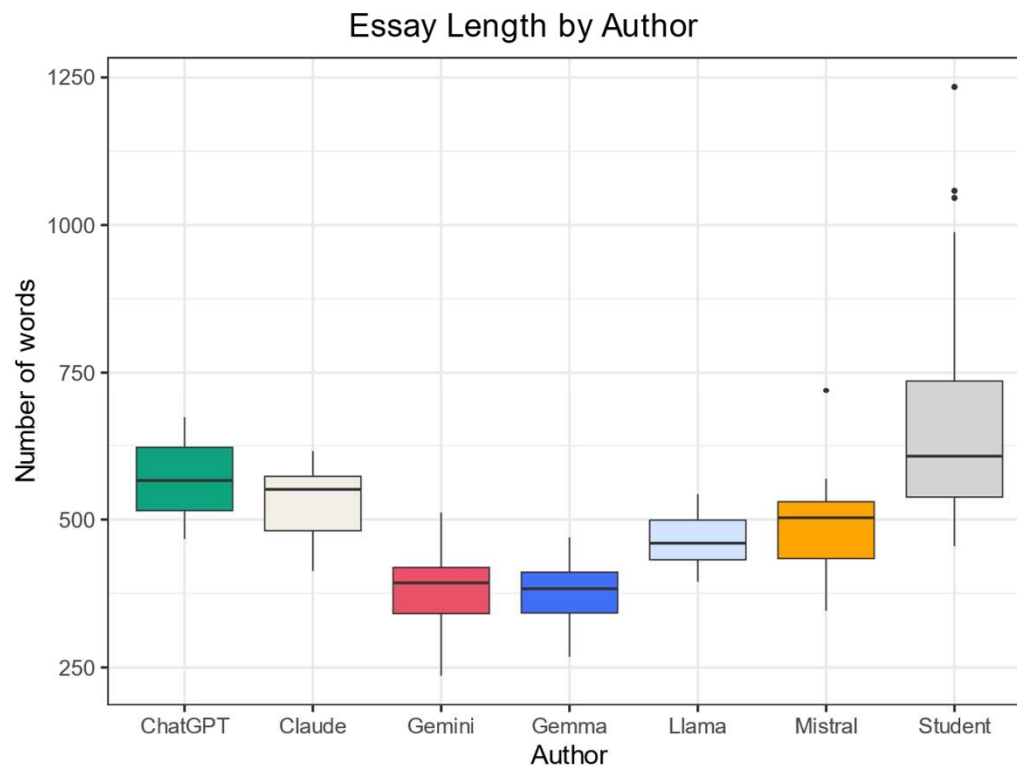
The error bar (one standard deviation) shows greater variability in essays by students.

ChatGPT writes longer essays than other LLMs, followed by Claude.

Gemini (close LLM) and Gemma (open) write shorter essays (both models are by Google).

Mistral shows more variability than other LLMs.

Essay length



The median essay length by students is very close to that of ChatGPT and Claude.

There is more variability in student essays (some outliers: few, very long texts).

LLMs are more consistent (especially ChatGPT (closed model) and Llama (open)).

Claude, Gemini, Mistral show more variability below the median.

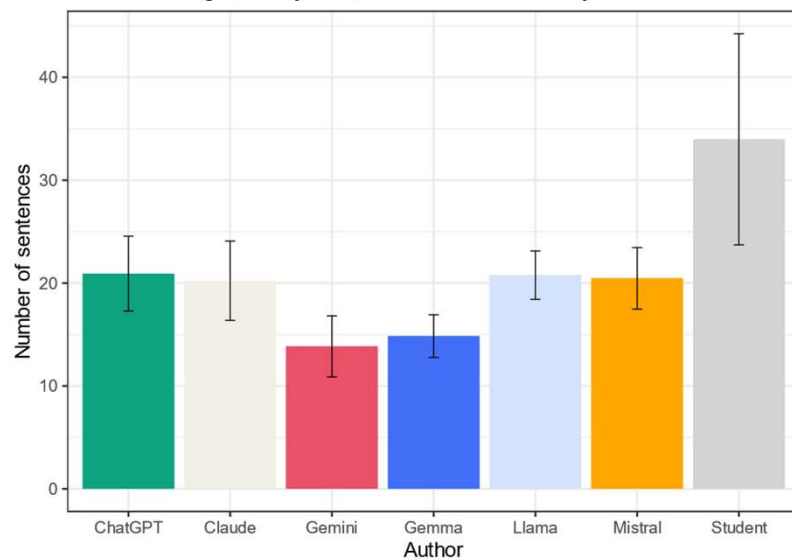
One outlier in Mistral (possibly due to instability in API calls); the value is not too extreme (z-score < 2.5), so the outlier was not removed.

Number of sentences and words per sentence

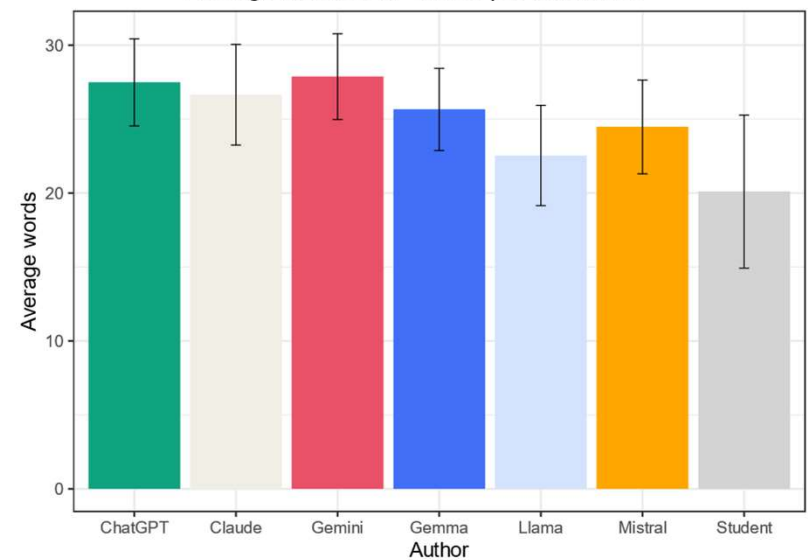
Student essays contain more sentences on average, but they show more variability.

The average number of word per sentence is higher for all language models than for students.

Average Essay Number of Sentences by Author



Average Number of Words per Sentence



Differences in the choice of adverbials

Category	Used by both Students and LLMs	Used only by Students	Used only by LLMs	Not used
Apposition	for example, for instance, altogether	that is, in other words	specifically, namely	that is to say
Contrast/Concession	yet, instead, however, rather, on the other hand, still, in contrast, though, of course	anyway	by contrast, conversely, nevertheless	anyhow, at any rate, in any case, nonetheless
Enumeration and Addition	furthermore, moreover, in the first place, too, secondly, to begin with, first, as well, thirdly, in addition, lastly, also, as well as, firstly, finally, additionally, second	besides, then, next	fourthly, similarly	for another thing, for one thing, fourth, in the fourth place, in the second place, in the third place, likewise, third, to start with, what's more

Differences in the choice of adverbials

Category	Used by both Students and LLMs	Used only by Students	Used only by LLMs	Not used
Result/ Inference	as a result, so, therefore, then, thus, consequently	accordingly	thereby	hence, in consequence, in that case
Summation	in conclusion	overall, to conclude, all in all		by the same token, in sum, in summary, to sum up, to summarize
Transition		now, by the way	meanwhile, in comparison	incidentally

Occurrences of adverbials for each category

	Apposition	Contrast/ Concession	Enumeration and Addition	Result/ Inference	Summation	Transition	Total
ChatGPT	14	22	74	18	13	1	142
Claude	7	29	47	2	12	1	98
Gemini	2	23	28	5	6	1	65
Gemma	4	11	28	2	7		52
Llama	12	21	68	5	13		119
Mistral	21	17	64	9	13		124
Student	42	159	291	185	41	7	725
Total	102	282	600	226	105	10	1325

Frequencies of adverbials for each category

	Apposition	Contrast/ Concession	Enumeration and Addition	Result/ Inference	Summation	Transition	Total
ChatGPT	1.90	2.98	10.02	2.44	1.76	0.14	19.23
Claude	1.02	4.22	6.83	0.29	1.74	0.15	14.25
Gemini	0.40	4.61	5.62	1.00	1.20	0.20	13.04
Gemma	0.81	2.23	5.69	0.41	1.42	0.00	10.56
Llama	1.99	3.49	11.30	0.83	2.16	0.00	19.77
Mistral	3.23	2.62	9.85	1.39	2.00	0.00	19.09
Student	0.83	3.13	5.72	3.64	0.81	0.14	14.26

Frequencies are per 1000 words

Occurrences and frequencies of linking adverbials

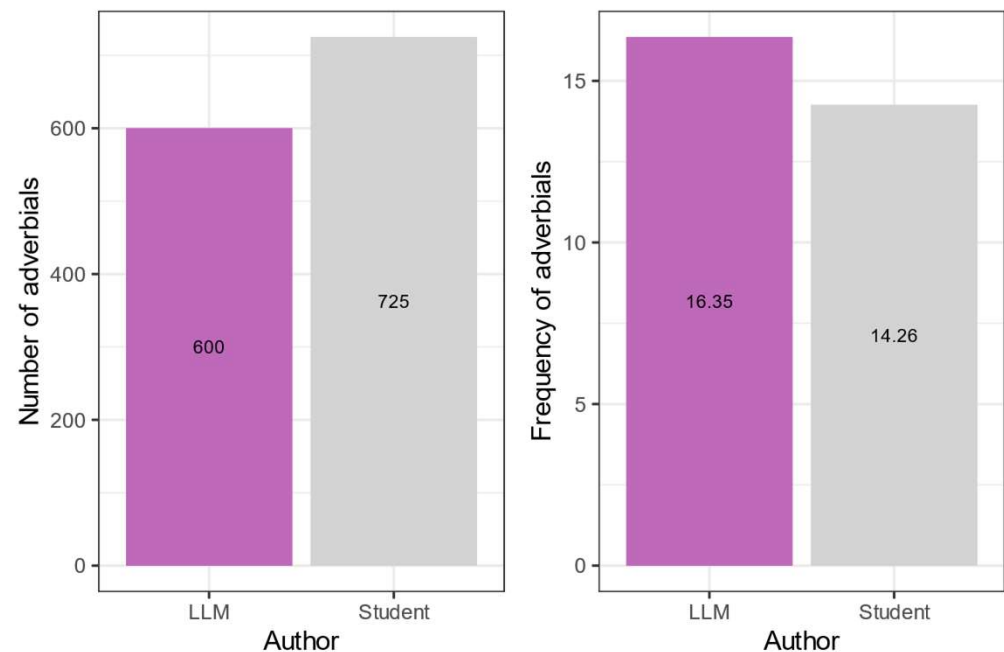
The number of adverbials in essays by students is higher, but linking adverbials are used more frequently by LLMs.

Here the frequency is computed as

$$\frac{\text{Number of Adverbials}}{\text{Number of Tokens}} * 1000$$

considering the essays by LLMs as a single corpus.

Adverbials by Author

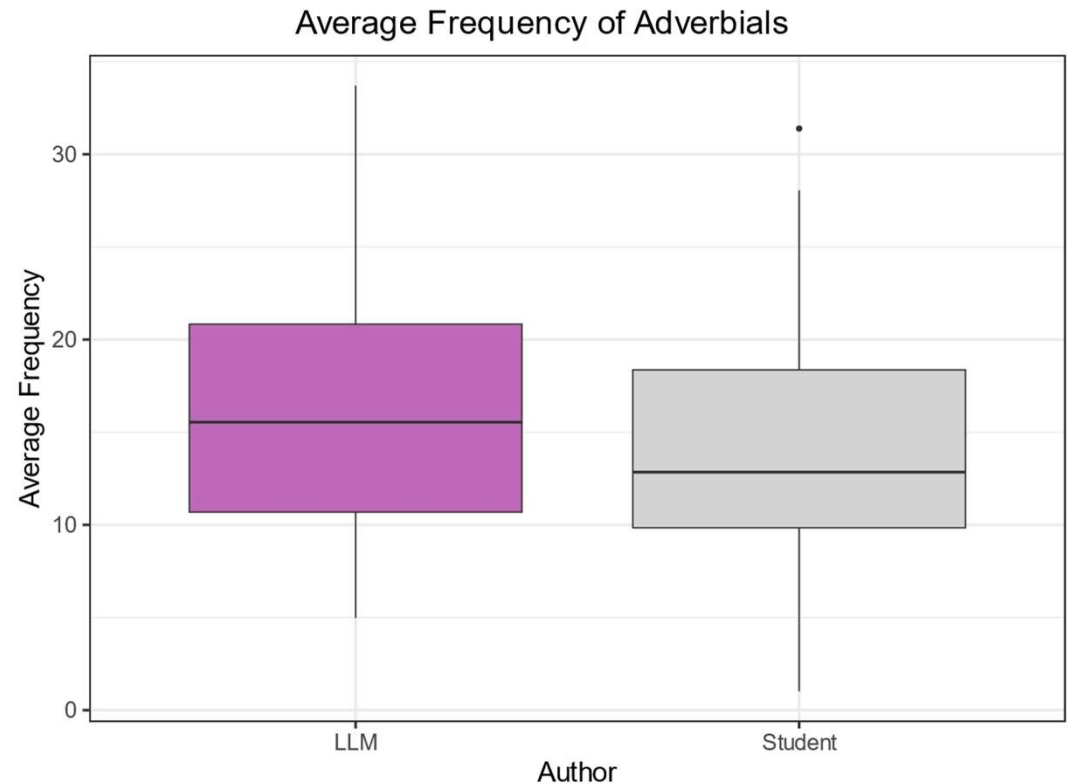


Occurrences and frequencies of linking adverbials

To see if the difference in frequencies is statistically significant, the independent two-tailed t-test was computed.

In this case, each file was considered as an observation and the frequency of adverbials computed for each file as the observed variable.*

The t-test showed that the difference is not statistically significant: $t(153.97) = 1.920$, $p\text{-value} = 0.057$.



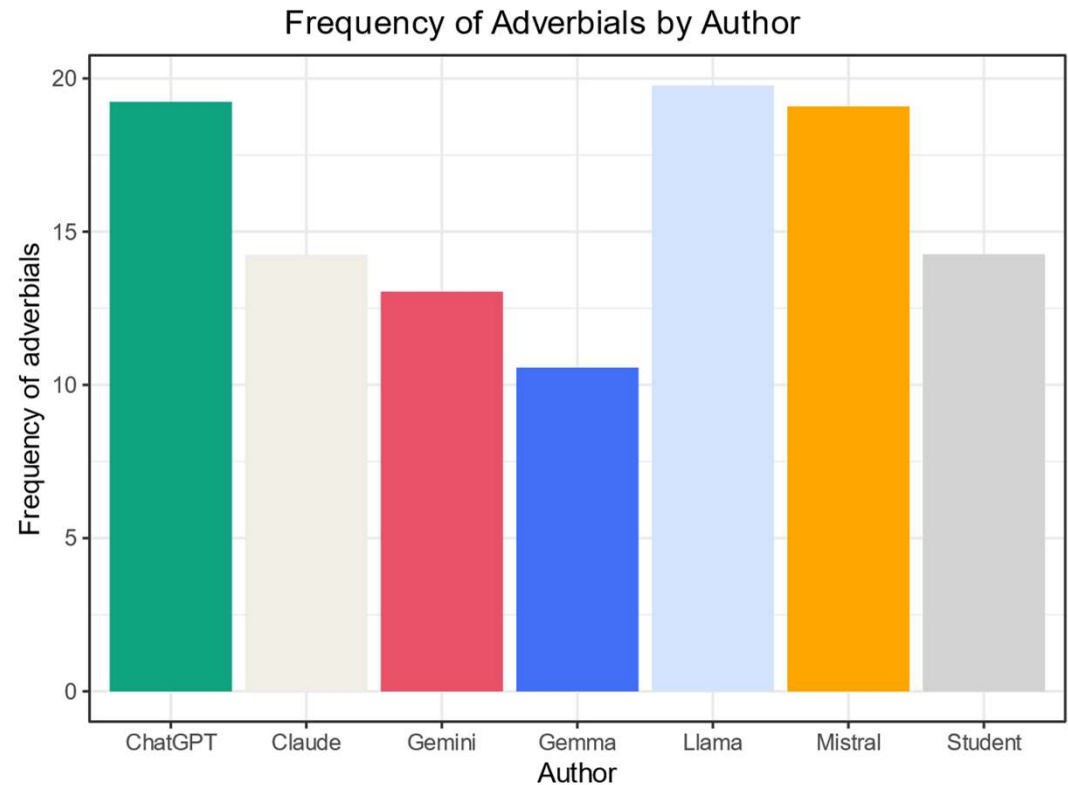
* There are 156 observations, so the t-test with Welch's adjustment can be applied even if the data is not normally distributed [Brezina 2018, p. 189]

Detail of frequency of adverbials

ChatGPT, Llama and Mistral use adverbials more frequently.

Claude is very close to students.

Gemini and Gemma show lower frequencies.



Statistical significance of differences in frequencies

The Dunn test was performed to see if there are statistically significant differences in the frequencies of adverbials between groups.

The Kruskal-Wallis test shows that there is a statistically significant difference for at least one group (p-value=0), so the Dunn test can be applied.

The Dunn test with Bonferroni correction shows that there are not statistically significant differences between Students and LLMs.

The differences are significant between Gemma and three other models: ChatGPT, Llama and Mistral.

```
> dunn.test(adv_by_file$adverbials_frequency, adv_by_file$model, method = "Bonferroni")
Kruskal-Wallis rank sum test
```

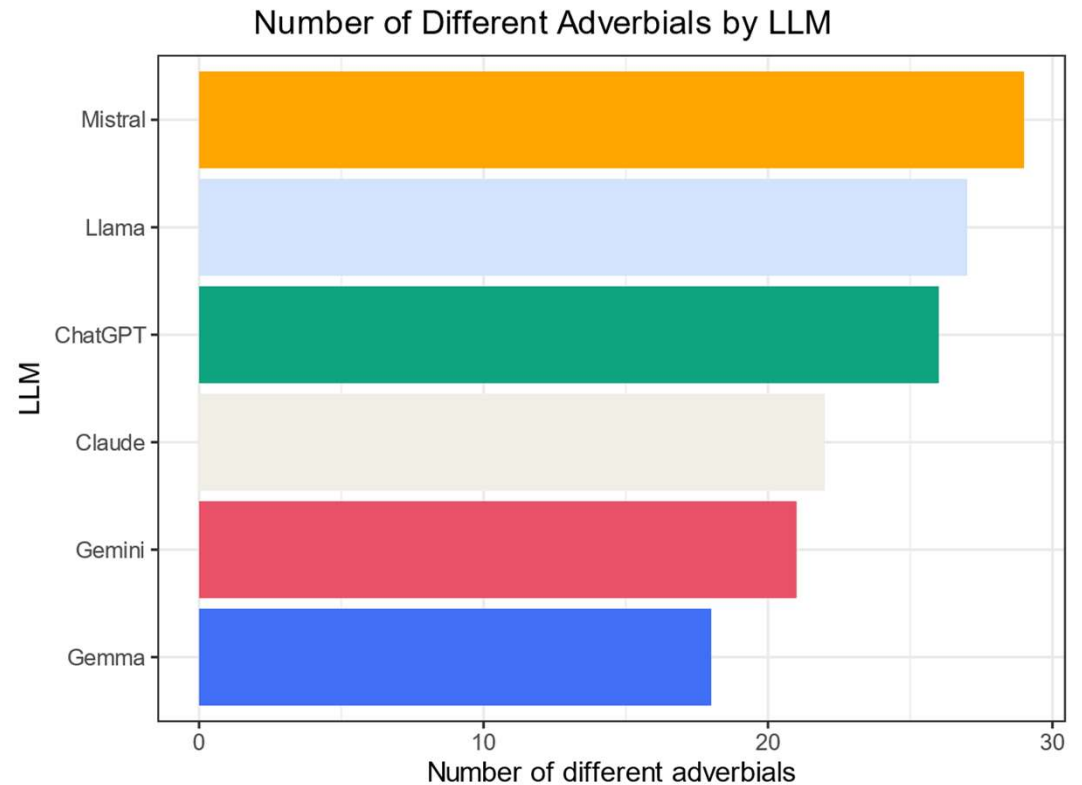
```
data: x and group
Kruskal-Wallis chi-squared = 26.6476, df = 6, p-value = 0
```

		Comparison of x by group (Bonferroni)					
Col Mean-	Row Mean	ChatGPT	Claude	Gemini	Gemma	Llama	Mistral
Claude	1.640906 1.0000						
Gemini	2.075009 0.3988	0.434102 1.0000					
Gemma	3.223210 0.0133*	1.582303 1.0000	1.148200 1.0000				
Llama	-0.510070 1.0000	-2.150977 0.3305	-2.585079 0.1022	-3.733280 0.0020*			
Mistral	-0.403715 1.0000	-2.044622 0.4294	-2.478724 0.1384	-3.626925 0.0030*	0.106355 1.0000		
Student	2.260231 0.2500	0.111780 1.0000	-0.456593 1.0000	-1.959941 0.5250	2.928070 0.0358	2.788818 0.0555	

```
alpha = 0.05
Reject Ho if p <= alpha/2
```

Variety of linking adverbials by LLM

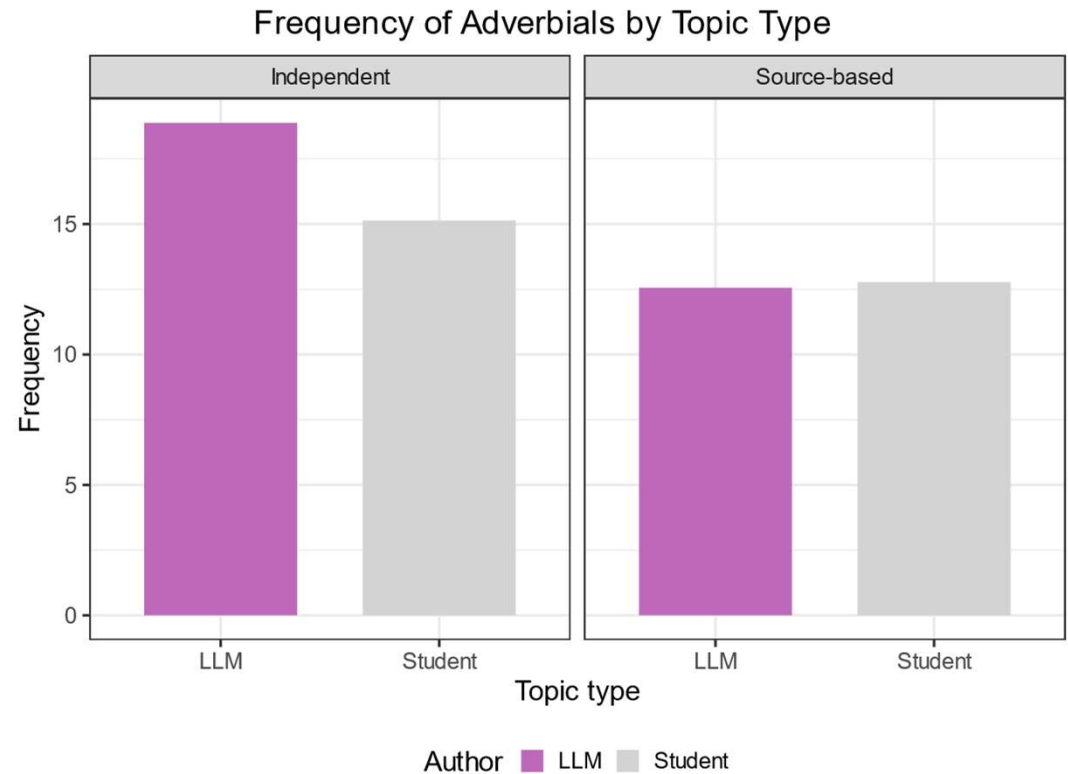
Mistral and Llama use a wider variety of different linking adverbials compared to others.



Frequency of adverbials by topic type

Both students and LLMs use adverbials more frequently in independent essays.

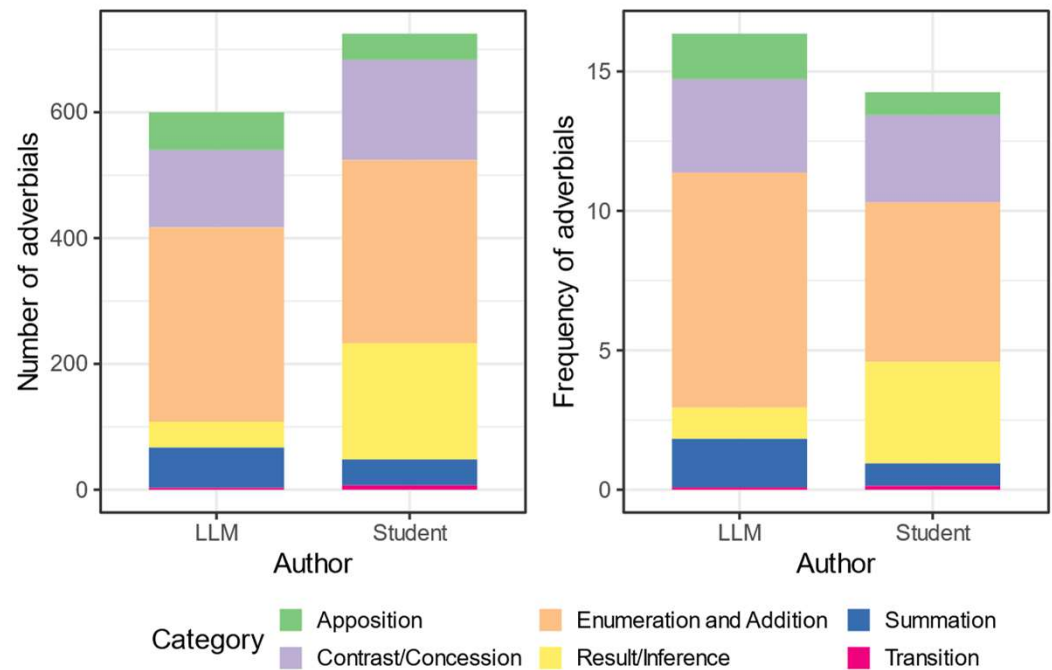
The frequencies in source-based essays are very similar.



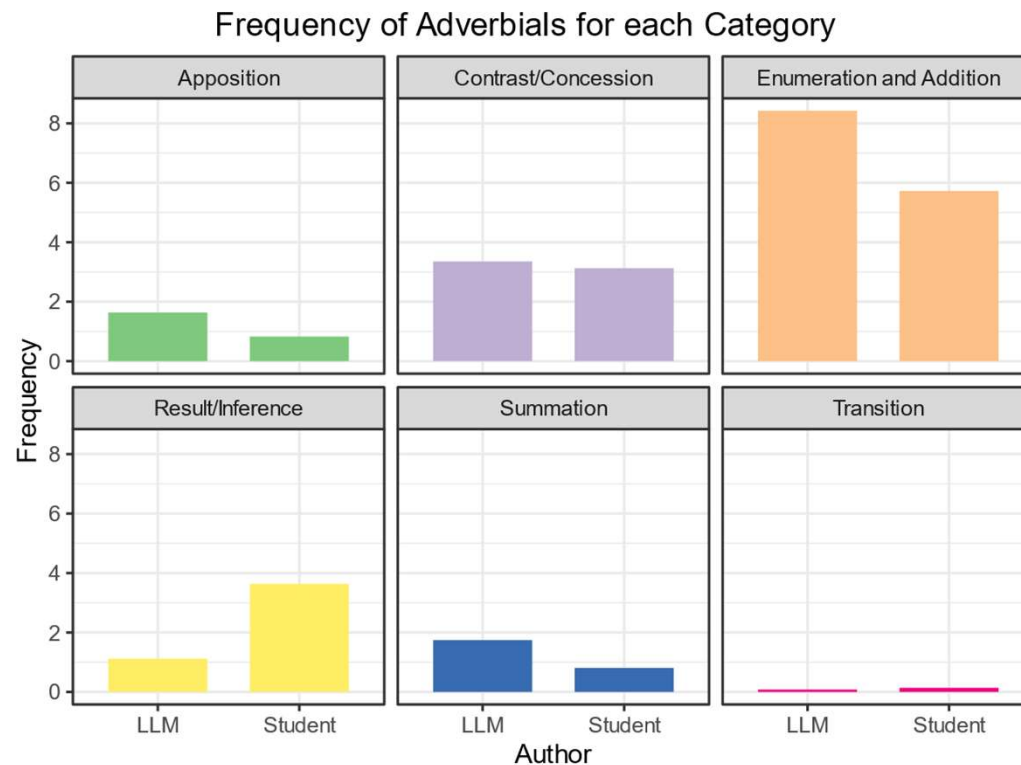
Occurrences and frequencies of Biber's categories

LLMs and Students show different proportions in the use of the functional categories from Biber's classification.

Adverbial Categories by Author



Frequency of Biber's categories



Apposition, Enumeration and Addition, and Summation adverbials are used more frequently by LLMs.

Result/Inference adverbials on the contrary are used more frequently by Students.

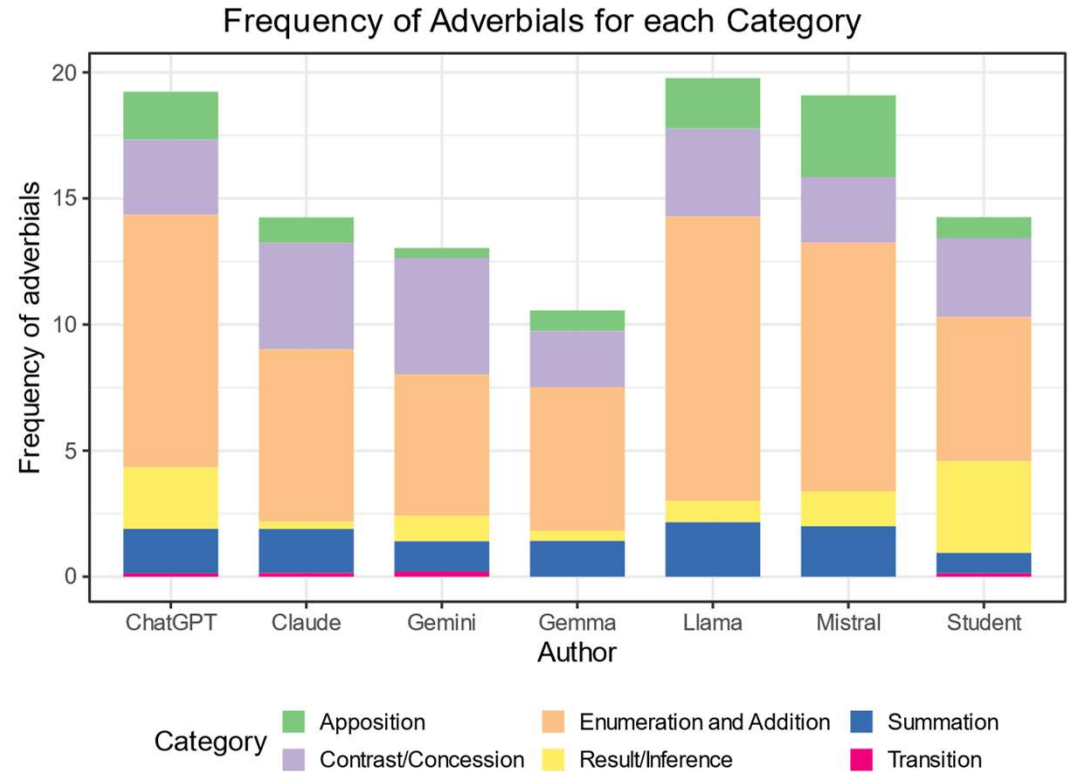
The frequencies of Contrast/Concession adverbials are very close.

Transition adverbials are very infrequent both in LLMs and Student essays.

Detail of Biber's categories

There are also differences across LLMs.

The differences seem to be relevant for the Apposition, Enumeration and Addition, Summation and Result/Inference categories, so the Kruskal-Wallis test was performed on these categories to see if the differences are statistically relevant.



```
[1] "Apposition"
> dunn.test(appos$construction_frequency, appos$model, method = "Bonferroni")
Kruskal-Wallis rank sum test

data: x and group
Kruskal-Wallis chi-squared = 24.3673, df = 6, p-value = 0
```

Comparison of x by group (Bonferroni)						
Col Mean- Row Mean	ChatGPT	Claude	Gemini	Gemma	Llama	Mistral
Claude	1.347971 1.0000					
Gemini	2.521384 0.1227	1.173413 1.0000				
Gemma	1.638900 1.0000	0.290929 1.0000	-0.882484 1.0000			
Llama	0.152737 1.0000	-1.195233 1.0000	-2.368647 0.1875	-1.486162 1.0000		
Mistral	-1.287360 1.0000	-2.635332 0.0883	-3.808745 0.0015*	-2.926261 0.0360	-1.440098 1.0000	
Student	2.226239 0.2730	0.461331 1.0000	-1.075028 1.0000	0.080415 1.0000	2.026258 0.4488	3.911790 0.0010*

alpha = 0.05
Reject Ho if p <= alpha/2

```
[1] "Result/Inference"
> dunn.test(result$construction_frequency, result$model, method = "bonferroni")
Kruskal-Wallis rank sum test

data: x and group
Kruskal-Wallis chi-squared = 43.7328, df = 6, p-value = 0
```

Comparison of x by group (Bonferroni)						
Col Mean- Row Mean	ChatGPT	Claude	Gemini	Gemma	Llama	Mistral
Claude	2.501424 0.1299					
Gemini	1.379163 1.0000	-1.122260 1.0000				
Gemma	2.451846 0.1492	-0.049577 1.0000	1.072682 1.0000			
Llama	1.870434 0.6449	-0.630989 1.0000	0.491270 1.0000	-0.581412 1.0000		
Mistral	1.221416 1.0000	-1.280008 1.0000	-0.157747 1.0000	-1.230430 1.0000	-0.649018 1.0000	
Student	-1.106463 1.0000	-4.381596 0.0001*	-2.912212 0.0377	-4.316684 0.0002*	-3.555437 0.0040*	-2.705672 0.0716

alpha = 0.05
Reject Ho if p <= alpha/2

```
[1] "Enumeration and Addition"
> dunn.test(enum$construction_frequency, enum$model, method = "Bonferroni")
Kruskal-Wallis rank sum test

data: x and group
Kruskal-Wallis chi-squared = 29.8844, df = 6, p-value = 0
```

Comparison of x by group (Bonferroni)						
Col Mean- Row Mean	ChatGPT	Claude	Gemini	Gemma	Llama	Mistral
Claude	1.176570 1.0000					
Gemini	2.105671 0.3699	0.929100 1.0000				
Gemma	1.975423 0.5063	0.798852 1.0000	-0.130247 1.0000			
Llama	-0.748924 1.0000	-1.925495 0.5688	-2.854595 0.0452	-2.724347 0.0677		
Mistral	-0.193200 1.0000	-1.369771 1.0000	-2.298871 0.2259	-2.168624 0.3162	0.555723 1.0000	
Student	3.043089 0.0246*	1.502596 1.0000	0.286118 1.0000	0.456652 1.0000	4.023661 0.0006*	3.296048 0.0103*

alpha = 0.05
Reject Ho if p <= alpha/2

```
[1] "Summation"
> dunn.test(sum$construction_frequency, sum$model, method = "Bonferroni")
Kruskal-Wallis rank sum test

data: x and group
Kruskal-Wallis chi-squared = 45.0601, df = 6, p-value = 0
```

Comparison of x by group (Bonferroni)						
Col Mean- Row Mean	ChatGPT	Claude	Gemini	Gemma	Llama	Mistral
Claude	-0.026742 1.0000					
Gemini	0.775530 1.0000	0.802272 1.0000				
Gemma	0.285252 1.0000	0.311994 1.0000	-0.490277 1.0000			
Llama	-1.704829 0.9264	-1.678086 0.9800	-2.480359 0.1378	-1.990081 0.4891		
Mistral	-1.089753 1.0000	-1.063011 1.0000	-1.865283 0.6525	-1.375006 1.0000	0.615075 1.0000	
Student	2.846836 0.0464	2.881850 0.0415	1.831429 0.7039	2.473353 0.1405	5.078981 0.0000*	4.273658 0.0002*

alpha = 0.05
Reject Ho if p <= alpha/2

Kruskal-Wallis and Dunn test on the categories

The Kruskal-Wallis test shows that there are a statistically significant differences for at least one group (p-value=0) in all categories, so the Dunn test can be performed.

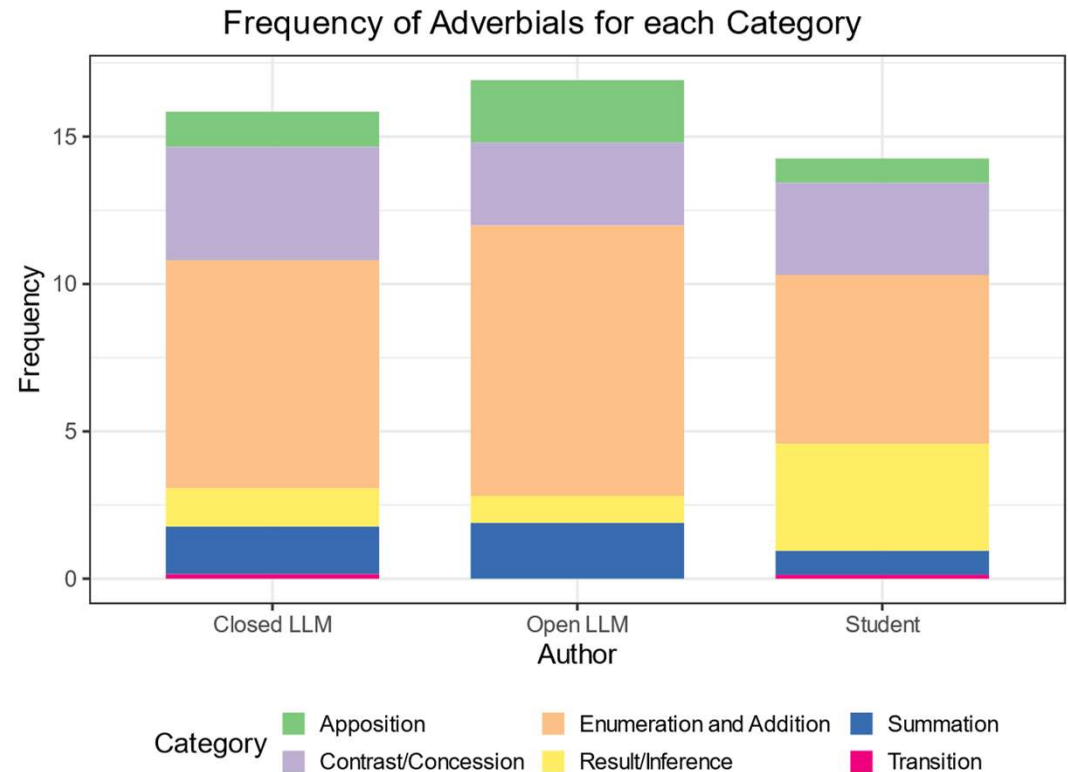
The Dunn test with Bonferroni correction shows that for the Enumeration and Addition, and Result/Inference categories the differences are statistically relevant between Students and three LLMs.

Comparison between Open LLMs, Closed LLMs and Students

Open LLMs use linking adverbials more frequently than Closed LLMs and Students.

The Kruskal-Wallis test showed that difference in frequencies between Open LLM, Closed LLMs and Students is not statistically relevant (chi-squared = 12.25, df = 6, p-value = 0.06).

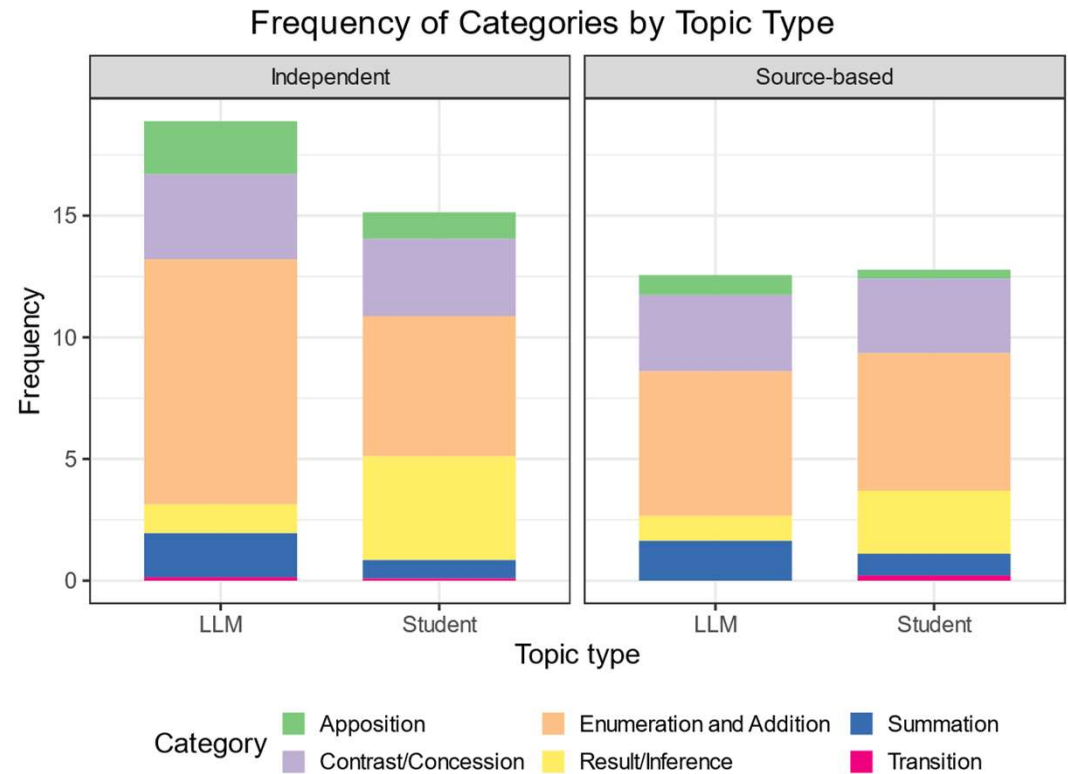
Open LLMs use even fewer Result/inference adverbials than Closed LLMs.



Frequency of categories by topic type

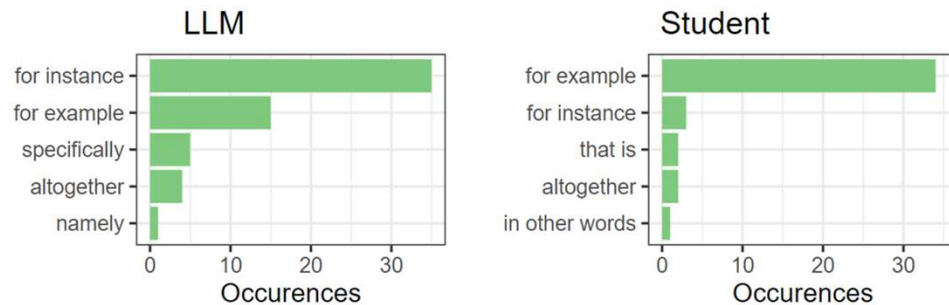
In independent essays LLMs use Enumeration and Addition adverbials very frequently, while Result/Inference adverbials are rare.

In source-based essays, both groups use fewer adverbials across all categories compared to independent essays. The reduction in Result/Inference adverbials is notable in LLMs



Most used Apposition and Summation adverbials

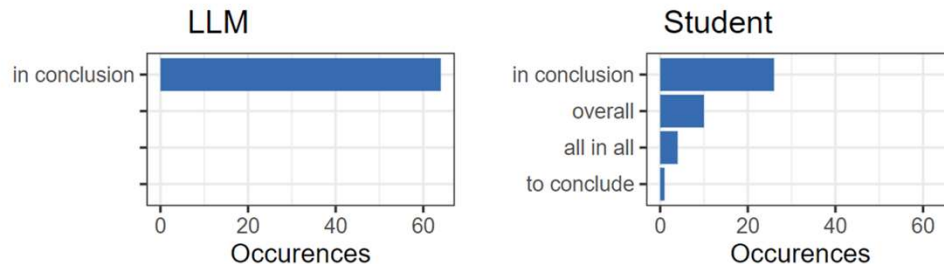
Apposition



For Apposition adverbials, “for instance” and “for example” are the most used by both LLMs and Students, but “for instance” occurs only sporadically in student essays.

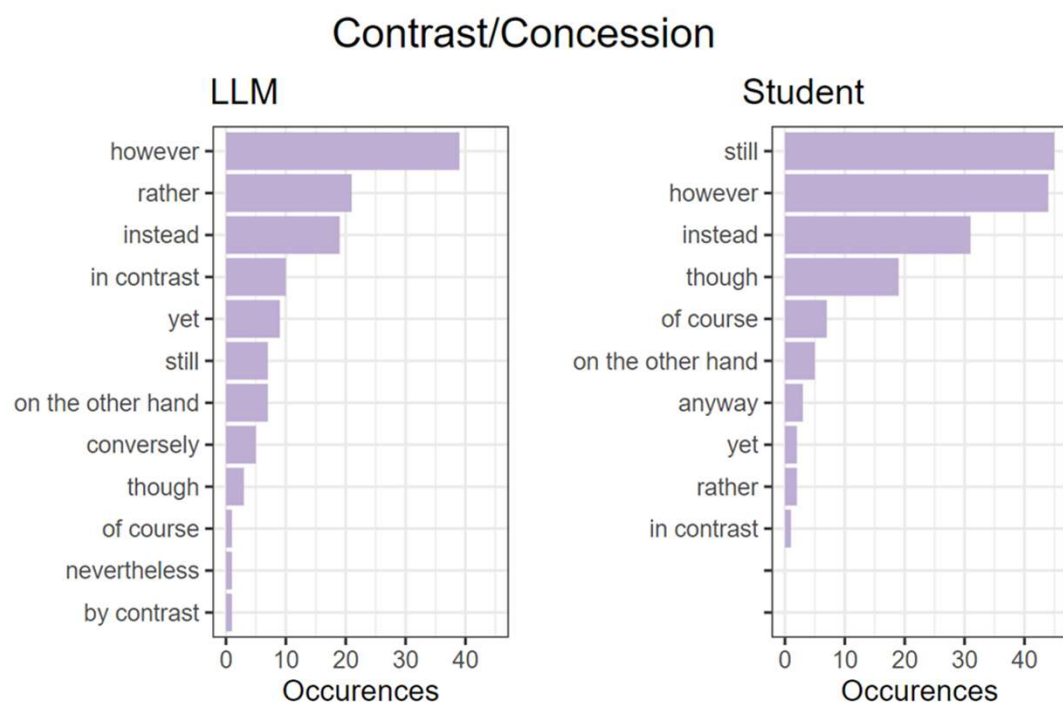
“Specifically” and “namely” are used only by LLMs while “that is” and “in other words” only by students.

Summation



For Summation adverbials, LLMs use only “in conclusion”, while students choose among four options.

Most used Contrast/Concession adverbials

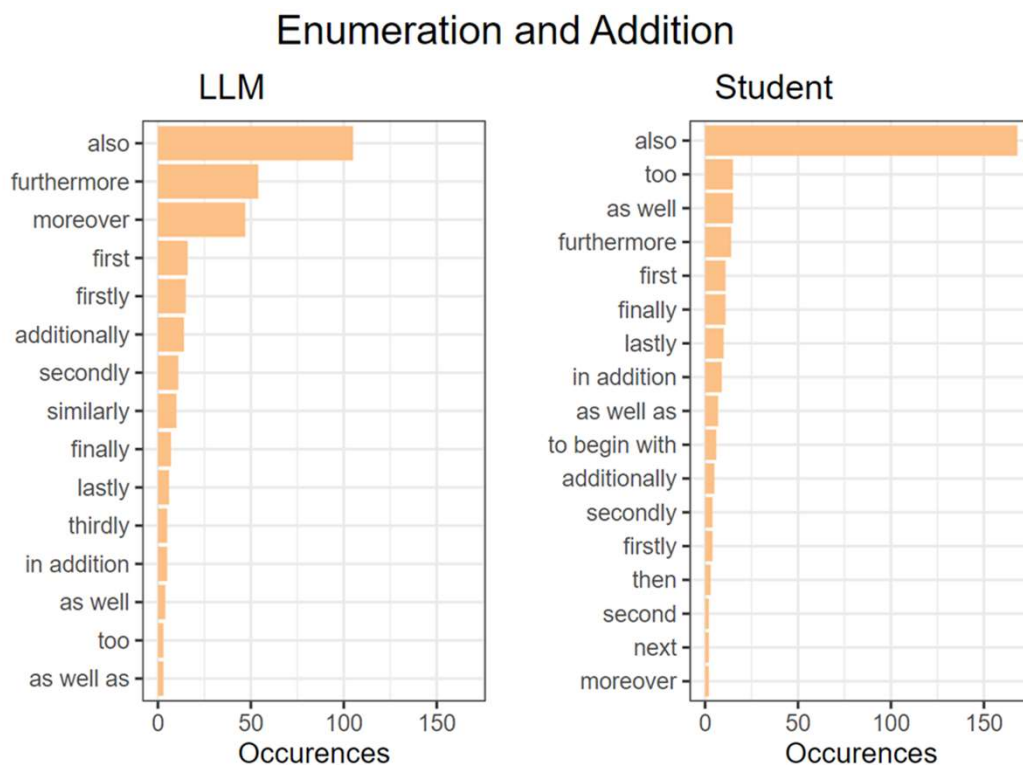


"still" is the preferred Contrast/Concession adverbial for students, while "however" is the most used by LLMs.

In this case, LLMs show more variability, they use "conversely", "nevertheless" which do not appear in the Student corpus.

Students rely more heavily on words like "of course" and "anyway".

Most used Enumeration and Addition adverbials



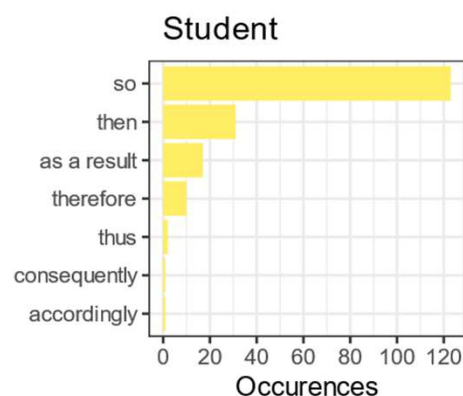
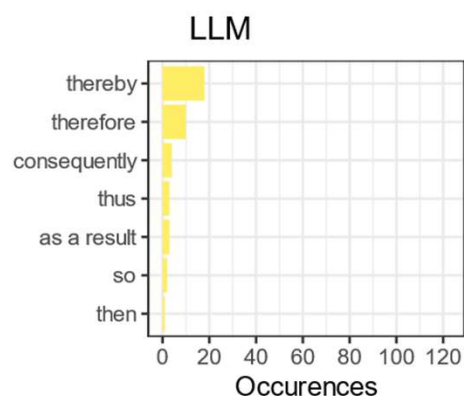
Both the LLMs and students use the word "also" far more frequently than any other adverbial, especially students.

Students frequently use "too", "as well", which are rare in essays by LLMs.

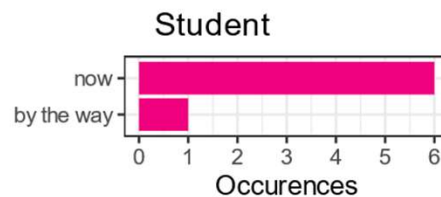
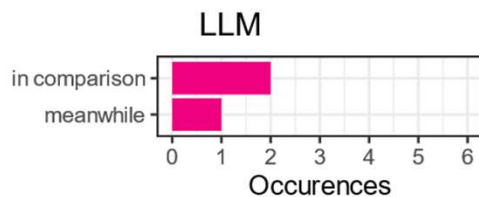
LLMs use "furthermore", "moreover", "additionally" and "similarly" more frequently than students.

Most used Result/Inference and Transition adverbials

Result/Inference



Transition



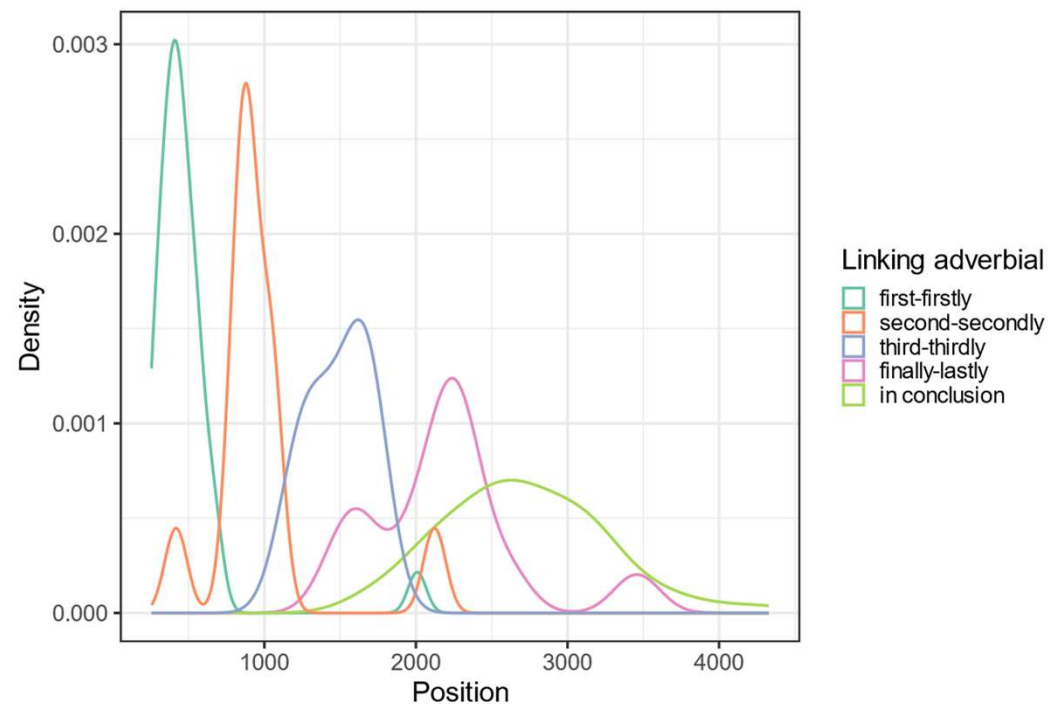
For Result/Inference adverbials, "so" is the most used by students, followed by "then", while LLMs prefer "thereby", which is not used by students, "therefore" and "consequently".

For Transition adverbials, LLMs and students make completely different choices: "in comparison" and "meanwhile" are used by LLMs and "now" and "by the way" by students.

Use of Enumeration and Addition adverbials in LLMs

The density plot of the positioning of the occurrences of linking adverbials shows that they are used appropriately by LLMs (occurrences of “first-firstly” appear at the beginning, followed by “second-secondly”...)

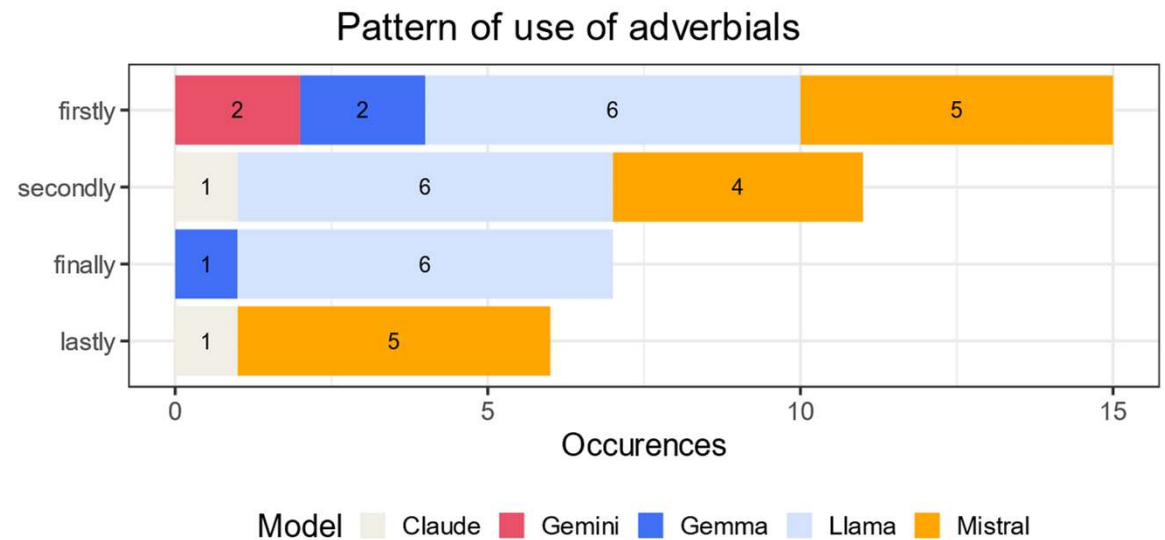
Positioning of adverbials by LLMs



Pattern of use of Enumeration and Addition adverbials in LLMs

Llama and Mistral frequently use “firstly” followed by “secondly”.

While Llama uses “finally” six times to conclude the enumeration and never uses “lastly”, Mistral uses only “lastly” five times.



Use of “First and foremost” in LLMs

The only collocate for “first” in LLMs corpus is “foremost”.

“First and foremost” is used frequently by LLMs but it doesn’t appear in the Student corpus.

The screenshot shows the AntConc interface with the 'Collocate' tab selected. The search query is 'first'. The results table shows one hit: 'foremost' with a likelihood of 7.515. The target corpus is 'argumentation_LL' with 78 files and 36592 tokens.

Collocate	Rank	FreqL	FreqR	Range	Likelihood	Effect
1 foremost	1	14	0	14	120.440	7.515

The screenshot shows the AntConc interface with the 'Collocate' tab selected. The search query is 'first'. The results table shows 14 hits, each with a left context, the hit 'foremost', and a right context. The target corpus is 'argumentation_LL' with 78 files and 36592 tokens.

File	Left Context	Hit	Right Context
1 ChatGPT_Ce...	he overall school environment. First and	foremost,	allowing students to have access to the
2 ChatGPT_Su...	sable skills in the 21st century. First and	foremost,	allowing students to design their own p
3 Llama_Cell ...	lass time, is the better option. First and	foremost,	allowing students to have phones durin
4 ChatGPT_Co...	l in our interconnected world. First and	foremost,	the act of participating in community s
5 Mistral_Cell ...	late my reasons for this belief. First and	foremost,	the integration of technology into our c
6 ChatGPT_Ph...	, and technological evolution. First and	foremost,	the primary objective of any road regul
7 Llama_Com...	ll students is an excellent idea. First and	foremost,	community service instills a sense of so
8 Mistral_Co...	my reasons for this conviction. First and	foremost,	engaging in community service provide
9 Mistral_Man...	a well-rounded student body. First and	foremost,	extracurricular activities play a pivotal r
10 ChatGPT_Gr...	onal system strives to achieve. First and	foremost,	it is essential to recognize that particip
11 ChatGPT_Di...	dualized learning experiences. First and	foremost,	one cannot overlook the significant adv
12 Llama_A ...	in invaluable life experiences. First and	foremost,	participating in the Seagoing Cowboys
13 Mistral_Sum...	ples to substantiate my claim. First and	foremost,	student-designed summer projects em
14 Claude_Cell ...	ion for our school community. First and	foremost,	this policy strikes a reasonable balance

Use of “not only...but also” in LLMs

The pattern “not only...but also” is used frequently by LLMs and sparingly by Students (6 occurrences).

KWIC

Plot

File View

Cluster

N-Gram

Collocate

Word

Keyword

Wordcloud

Total Hits: 73

Page Size

100 hits

1 to 73 of 73 hits

	File	Left Context	Hit	Right Context
1	ChatGPT_Ce...	his policy would bring to our school community, as it prepares students not only for academic success	but also	for their roles as responsible digital citizens in a rapidly evolving world. Thank you
2	Mistral_Man...	e extracurricular involvement is not only beneficial for the students' personal and social development	but also	for their academic success. In conclusion, the principal's decision to require all stud
3	Gemma_Su...	outlining methodologies, and anticipating challenges – skills essential not only for academic success	but also	for navigating the complexities of real-world endeavors. A student tasked with des
4	Gemini_Cell ...	right prohibition, we can create an environment that prepares students not only for academic success	but also	for responsible and ethical participation in our increasingly digital world. Thank yo
5	ChatGPT_Su...	e freedom to shape their learning experiences, educators not only equip them with essential life skills	but also	foster a lifelong passion for learning that extends well beyond the classroom. Ther
6	Llama_Sum...	act of plastic waste on local wildlife. This project would not only engage them in meaningful learning	but also	foster a sense of purpose and responsibility. Secondly, student-designed projects e
7	ChatGPT_Ca...	Such reductions not only contribute to improved air quality and decreased greenhouse gas emissions	but also	foster community engagement and enhance the overall quality of life in urban area
8	Claude_Sum...	alysis and community outreach. This self-directed endeavor would not only reinforce academic skills	but also	foster real-world problem-solving abilities and a sense of civic engagement. In cor
9	ChatGPT_Ca...	or public transportation. The widespread participation in this initiative not only alleviates air pollution	but also	fosters a sense of community and encourages a cultural shift towards more sustain
10	Claude_Car...	impressive 70% of families living car-free. This approach not only reduces greenhouse gas emissions	but also	fosters a more compact and accessible community, demonstrating that suburban a
11	ChatGPT_D...	as the need for runoff elections. This structure not only preserves the integrity of the electoral process	but also	fosters confidence among the electorate that their votes contribute meaningfully tr
12	ChatGPT_Se...	ly, this multifaceted approach to decision-making not only enhances the quality of the choices made	but also	fosters personal growth and greater relational connections with others.
13	Claude_Man...	red outcomes without resorting to coercion. This approach would not only respect student autonomy	but also	cultivate a genuine love for extracurricular activities that extends far beyond the hig
14	ChatGPT_Se...	eself to a variety of perspectives, individuals not only enrich their understanding of the issues at stake	but also	cultivate critical thinking skills that are essential for navigating the complexities of
15	Gemini_Su...	bracing the power of student-designed projects, we not only foster deeper learning and engagement	but also	cultivate essential life skills that empower students to become self-directed, lifelon
16	ChatGPT_Se...	anced understanding of their options. This process not only enriches the decision-making experience	but also	cultivates a sense of intellectual humility, as one recognizes the limitations of their
17	Claude_See...	This process of active engagement with different ideas not only enriches the decision-making process	but also	cultivates a mindset of openness and adaptability, qualities that are increasingly cr
18	ChatGPT_Co...	e, as engaging in such activities not only fosters a sense of responsibility and empathy toward others,	but also	cultivates essential life skills that are increasingly vital in our interconnected world.
19	Gemma_Gra...	is educators should be to nurture well-rounded individuals who are not solely defined by their grades	but also	by their diverse talents, passions, and contributions to the school community. By rei

Search Query

Words

Case

Regex

Results Set

All hits

Context Size

20 token(s)

but also

Start

Adv Search

Sort Options

Sort to right

Sort 1 1R

Sort 2 2R

Sort 3 3R

Order by freq



Discussion



Writing style and usage of adverbials (Q1)

Writing style: essays by students are longer and contain more sentences, but essays by LLMs contain longer sentences, which indicates more sophisticated and complex language use in LLMs.

Usage of adverbials: LLMs use linking adverbials more frequently than students, but the difference has been shown not to be statistically relevant. The slight difference also shows that LLMs do not overuse linking adverbials to sound more authoritative. The difference is even smaller in source-based essays: this can be explained considering that independent essays demand more explicit connections between ideas, while source-based essays rely more on content derived from external sources for coherence.

Usage across categories: Enumeration and Addition is the most frequent category in both LLMs and students, showing that both groups tend to structure arguments by adding points in a list-like manner. LLMs show a preference for Apposition, Enumeration and Addition, and Summation adverbials, while students more frequently employ Result/Inference adverbials.

Formality and formulaic language (Q1)

Language formality: while there is a large overlap in the choice of adverbials, LLMs tend to use more formal adverbials (e.g., "for instance" instead of "for example", "specifically", "namely", "conversely", "nevertheless"...), while students use more colloquial language ("that is", "in other words", "still", "anyway", "of course", "too"...). In particular, for Result/Inference adverbials students use frequently "so" and "then", which Biber [Biber et al., 1998, p. 887] reports as the most frequent in conversation.

Formulaic language use: LLMs show a preference for fixed constructions ("Not only...but also", "First and foremost"), which are infrequent or absent in student corpus.

Repetitive patterns: Open-source LLMs (Llama, Mistral) show a repetitive use of certain patterns ("firstly", "secondly", "finally/lastly"), which results in more standardized, less diverse writing style. The use of formulaic expression and repetitive patterns reveals the statistical nature of LLMs training process.

Comparison between open and closed LLMs (Q2)

Gemma (the smallest open LLM) produces shorter essays and shows lower frequency of adverbial use, but it is comparable with Gemini (closed).

Closed LLMs (especially ChatGPT) are more consistent in essay length, while open LLMs (especially Mistral) show more variability. Claude (closed) shows some variability, similar to open models

Open models like Llama and Mistral perform comparably to closed models (e.g., ChatGPT) in terms of linking adverbial use and essay structure.

Open LLM (Llama and Mistral) rely more heavily on formulaic expressions (e.g., "firstly", "secondly", "finally"/"lastly").

ChatGPT is closer to students in the use of Result/Inference adverbials.

We can conclude that the larger open LLMs (Llama and Mistral) perform very closely to ChatGPT in argumentative writing and outperform other closed LLMs (especially Gemini).

Difference in Result/Inference adverbials (Q3)

The use of fewer Result/Inference adverbials by LLMs is worth analyzing.

[Herbold et al. 2023] found fewer discourse markers in AI-generated texts. According to them, the reason students use more discourse markers than LLMs “could be due to a greater reliance on explicit markers to guide readers through their arguments, whereas LLMs might generate text with more inherent logical flow, reducing the need for such markers”.

Here we propose another explanation based on Hyland’s theory of metadiscourse [Hyland, 2018].

Hyland makes a distinction between transition and structuring signals (which include Enumeration and Addition adverbials) and resultative and framing signals (which include Result/Inference adverbials).

Difference in Result/Inference adverbials (Q3)

Transition and structuring signals, “while essential for readability and flow, do not add any propositional information to a discourse but serve to make explicit relations that already exist in a text by virtue of the meanings of the propositions that they link”. They do not contribute to shaping the content of the text but serve to guide the reader through the argument.

In contrast, resultative and framing signals “are tied directly to the core content and argument of the text, often indicating critical connections and conclusions”. They show that the writer is actively guiding the reader toward a particular interpretation or conclusion, they are used to emphasize the importance of certain claims or to signal key turning points in the argument. They play a direct role in constructing and framing the argument within a specific perspective.

The lower frequency in LLM-generated essays indicates that LLMs are less inclined to express a sense of personal perspectives and persuasive interactions that are essential in argumentative writing.

Difference in Result/Inference adverbials (Q3)

The use of fewer adverbials suggests that LLMs focus more on logical flow than persuasion. Their emphasis is on how to argue, not on what is being argued.

This makes LLMs similar to ancient Greek Sophists in their use of rhetoric, where structure is prioritized over truth or conviction. Like the Sophists, LLMs can argue both sides without having personal beliefs or preferences.

As a result, LLM-generated arguments may feel mechanical or detached, lacking the personal touch that human writers use to persuade and express judgment.

This suggests investigating Hyland's interactional metadiscourse linguistic features to better understand the differences between LLMs and students in engaging with their readers, guiding interpretation, and managing interactions.



Conclusion



Key points

Linguistic Sophistication: LLMs demonstrate a capacity for producing sophisticated, formally structured argumentative essays, often using more complex sentence structures and formal language than students.

Adverbial Usage Patterns: While LLMs use linking adverbials slightly more frequently than students, there are notable differences in usage across categories. LLMs show a preference for Enumeration/Addition and Apposition adverbials, while students more frequently employ Result/Inference adverbials.

Formulaic vs. Natural Language: LLMs, particularly open-source models, show a tendency towards formulaic language patterns, which contrasts with the more varied writing style of students.

Personal Perspective: The lower frequency of Result/Inference adverbials in LLM-generated texts suggests potential limitations in injecting personal perspective into arguments, a key aspect of persuasive writing.

Further Research: Investigating Hyland's interactional metadiscourse to better understand the differences on how LLMs and humans engage with the readers.

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